INCH POUND
MIL-PRF-55339/48B
w/AMENDMENT 2
28 February 2023
SUPERSEDING
MIL-PRF-55339/48B
w/AMENDMENT 1
15 June 2017

PERFORMANCE SPECIFICATION

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY, (BETWEEN SERIES SMA TO SERIES TNC), CLASS 2, STRAIGHT PLUG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55339.

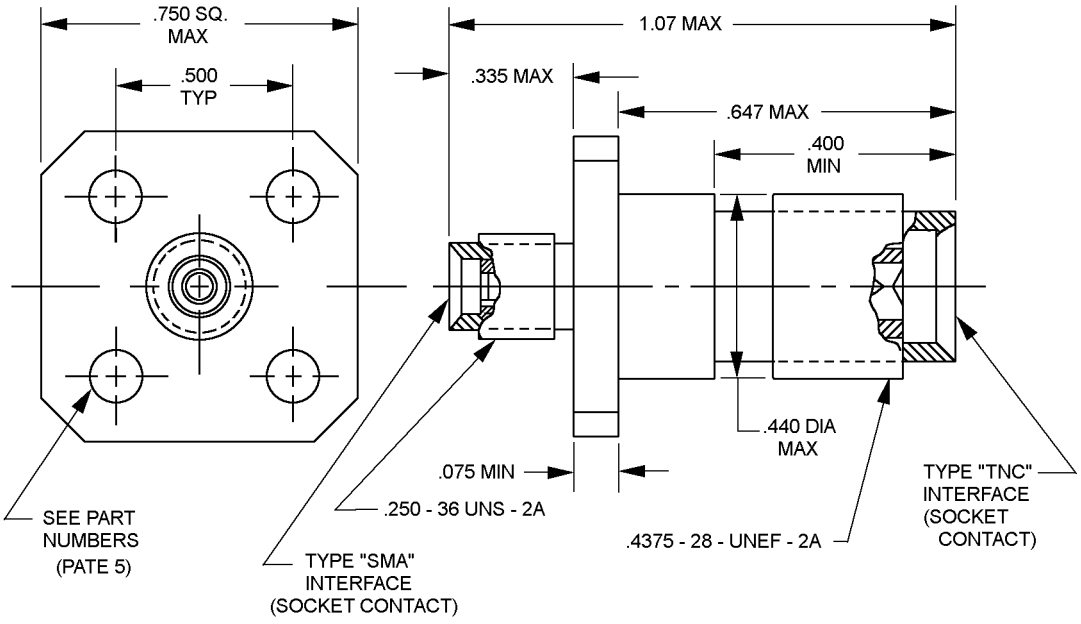
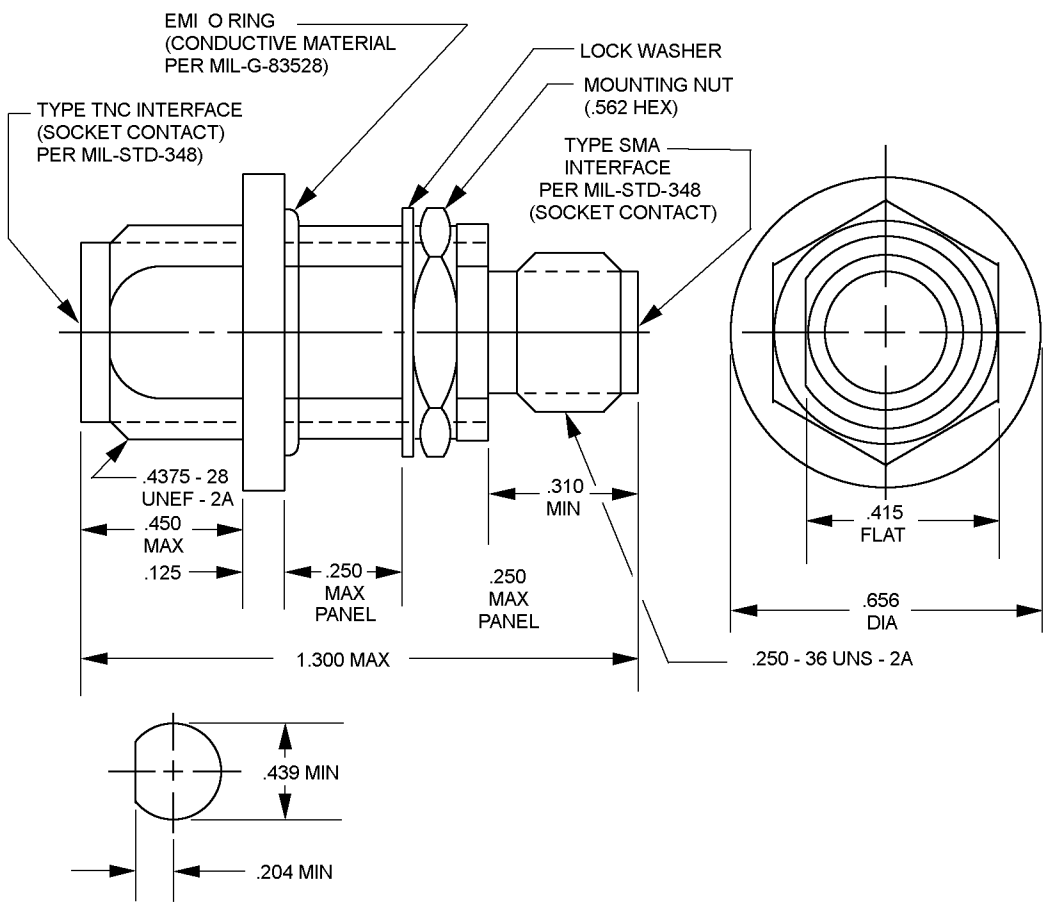


FIGURE 1. General configuration.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 5935

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RECOMMENDED MOUNTING HOLE

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .015.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Interfaces shall be in accordance with MIL-STD-348.

FIGURE 1. General configuration – Continued.

ENGINEERING DATA:

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Nominal impedance: 50 ohms.

Frequency range: 0 to 15 GHz.

Voltage rating:

335 V rms at sea level.

85 V rms at 70,000 feet.

Operating temperature range: -65°C to +165°C.

Performance: Installation torque 7 to 10 inch-pounds, series SMA; 4 to 6 inch-pounds series TNC.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Center contact retention:	Series SMA	Series TNC
Axial force (pounds, minimum)	6	6
Torque	Not applicable	Not applicable
Force to engage and disengage:	**Series SMA**	**Series TNC**
Longitudinal force (pounds, minimum)	Not applicable	Not applicable
Torque (inch-pounds, maximum)	2	2
Mating characteristics:	**Series SMA**	**Series TNC**
Center contact (socket):		
Oversize test pin diameter (inch)	.0375 + .0001	.057 diameter minimum (nonclosed entry contacts only)
Insertion depth (inch, minimum)	.030/.045	.125
Number of insertions	3	1
Pin finish (microinches)	16	--
Maximum test pin (insertion force test):		
Steel test pin diameter (inch, minimum)	.0370 + .0001	.054
Pin finish (microinches)	16	16
Insertion force (pounds, maximum)	3	2
Insertion depth (inch, minimum)	.050/.075	--

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Minimum test pin (withdrawal force):	Series SMA	Series TNC
Steel test pin diameter (inch)	.0355 - .0001	.052
Pin finish (microinches)	16	16
Withdrawal force (ounces, minimum)	1	2
Insertion depth (inch, minimum)	.050/.075	--

Permeability: Less than 2.0, air = 1.0.

Seal:

Pressurized: Not applicable.

Weatherproof: Not applicable.

Insulation resistance: 5,000 megohms, minimum.

VSWR:

1.05 + .008 F (GHz) at .5 to 12.4 GHz.

1.09 + .009 F (GHz) at 12.4 to 15 GHz.

RF leakage (total): -65 dB, minimum, 2 to 3 GHz.

RF insertion loss: .18 dB, maximum, at 9 GHz.

Durability: 500 cycles minimum at 12 cycles per minute, maximum, the adapter shall meet the mating characteristics and force to engage disengage requirements.

Dielectric withstanding voltage:

Test voltage: 1,000 V rms, minimum (sea level).

Contact resistance (milliohms, maximum):

Contact	Initial	After environmental
Center	4.1 1/	6.0
Outer	2.2	Not applicable

1/ Two center contacts must be mated to the center conductor under test, therefore doubling “center contact” resistance.

Vibration, high frequency:

Interruptions: 1 µs, maximum, test condition D.

Shock: MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-107, test condition C.

Moisture resistance: 200 megohms, minimum, within 5 minutes after removal from humidity.

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Salt spray: MIL-STD-202-101, test condition B.

Corona level:

Voltage: 250 V, minimum.

Altitude, 70,000 feet, minimum.

RF high potential withstanding voltage:

RF voltage: 1,000 V rms, minimum.

Frequency: 5 MHz, minimum.

Group qualification: See table I.

Part or identifying numbers (PIN): Also see table II.

M55339/48-00002 (for .125 inch diameter through holes).
M55339/48-00003 (tapped holes to accommodate 3-56 UNF mounting screws).
M55339/48-30001 (for .125 inch diameter through holes).
M55339/48-30002 (tapped holes to accommodate 3-56 UNF mounting screws).
M55339/48-00004 (jam-nut mounted).
M55339/48-30003 (jam-nut mounted).

Marking: As specified in MIL-PRF-55339.

TABLE I. Group qualification.

Group	Submission and qualification of any of the following adapters	Qualifies the following adapters
1	M55339/48-00002 M55339/48-00003 M55339/48-00004	M55339/48-00002 M55339/48-00003 M55339/48-00004
2	M55339/48-30001 M55339/48-30002 M55339/48-30003	M55339/48-30001 M55339/48-30002 M55339/48-30003

NOTE: When a QPL source is obtained, DESC drawing 85018 will be canceled.

TABLE II. Cross reference of PIN.

Preferred PIN	Superseded PIN
M55339/48-00002	M55339/48-00001

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: In addition to MIL-PRF-55339, this document references the following:

MIL-STD-202-101, MIL-STD-202-107, MIL-STD-202-213, MIL-STD-348
MIL-DTL-83528

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CONCLUDING MATERIAL

Custodians:
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2023-025)

Review activities:
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of information above using the ASSIST Online database at https://assist.dla.mil.

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